📃 Lab Manual: Practical 3

Course Code: ETCS406

Course Title: Game Development

Program: B. Tech Computer Science and Engineering

Semester: VII

Practical Title: Develop a Simple 2D Platformer in Unity

Duration: 4 Hours

Lab Type: Development & Implementation

Assessment Weightage: Part of CAL 1 (Lab Assignments)

Objective

To design and implement a basic 2D platformer game using Unity, incorporating sprites, tilemaps, and fundamental C# scripting for player movement and interaction.

Pre-requisites

- Familiarity with Unity Editor and interface
- Basic knowledge of C# scripting
- Understanding of sprites, tilemaps, and game physics
- Exposure to platformer mechanics (jumping, collision, gravity)

Learning Outcomes

By the end of this lab, students will be able to:

- Create a 2D game environment using Unity's tilemap system
- Implement player movement and jumping using C# scripts
- Integrate sprites and animations for visual feedback
- Apply basic collision detection and physics components

Tools & Resources

- Unity Hub and Unity Editor (2020.3 LTS or later recommended)
- Visual Studio or any C# IDE

- Free assets from Unity Asset Store or custom sprites
- Reference: Unity Learn – 2D Platformer Tutorial

Lab Tasks

Task 1: Project Setup

- Create a new 2D Unity project
- Set up the main camera and game resolution
- Import necessary assets (sprites, tiles, backgrounds)

Task 2: Environment Design

- Use Unity's Tilemap system to design a basic level
- Add platforms, obstacles, and background elements
- Configure colliders for tilemap and static objects

Task 3: Player Character Implementation

- Create a player GameObject with sprite and Rigidbody2D
- Add BoxCollider2D for collision detection
- Write a C# script for player movement (left/right) and jumping
- Implement gravity and ground detection logic

Task 4: Gameplay Features

- Add collectible items or enemies (optional)
- Implement simple UI (score, lives)
- Add sound effects or background music (optional)

Task 5: Testing & Debugging

- Playtest the game in Unity Editor
- Debug movement, collisions, and camera behavior
- Polish visuals and fix any gameplay bugs

Evaluation Criteria

Component	**Marks**
Level Design & Tilemap Usage	3

Component	Marks
Player Movement & Controls	3
Collision & Physics Setup	2
Code Quality & Functionality	2
Total	**10**

Tips for Success

- Keep the level short and focused for testing
- Use comments in your C# scripts for clarity
- Test edge cases like jumping on corners or falling off platforms
- Save your project frequently and use version control if possible

Suggested Reading & Resources

- *Unity Game Development Essentials* by Will Goldstone
- Unity Learn: Create with Code – 2D Platformer
- Brackeys YouTube: 2D Movement in Unity

Submission Guidelines:

- Submit your report in PDF format via the Google Classroom.
- Include screenshots or gameplay references where relevant.
- Submit a video of your game play.